higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 3

21 NOVEMBER 2019

This marking guideline consists of 5 pages.

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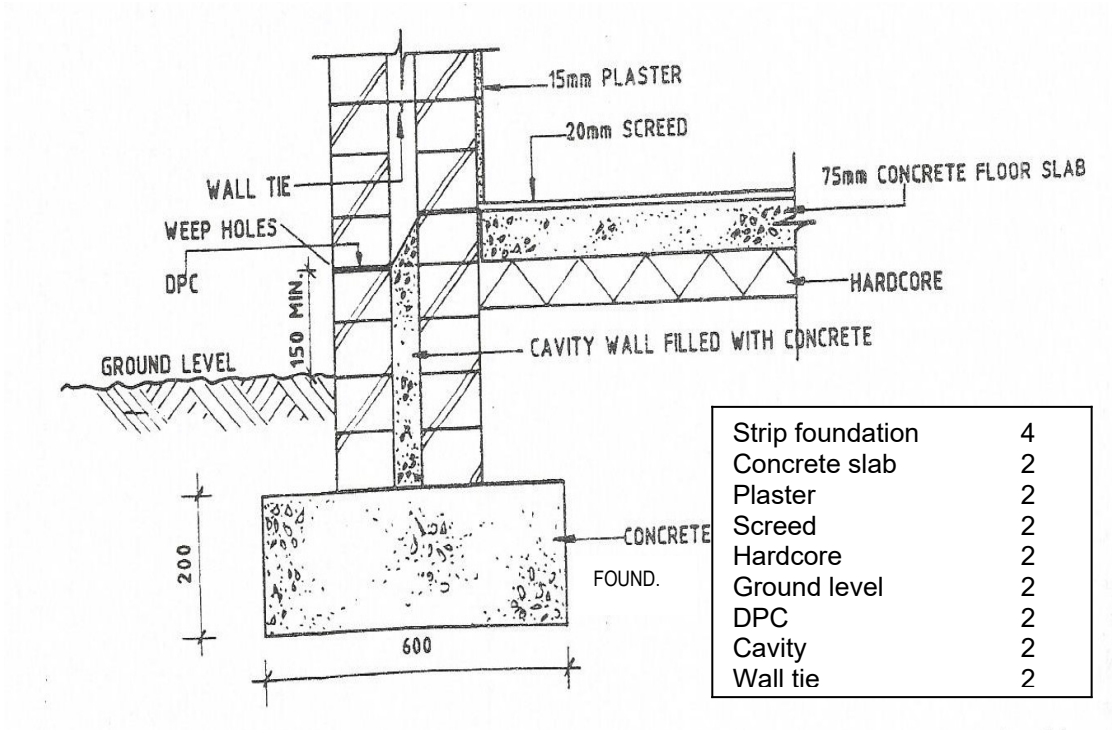
QUESTION 1: GENERAL

1.1 C
1.2 D
1.3 A
1.4 C
1.5 D
1.6 A
1.7 C
1.8 D
1.9 C
1.10 D

(10 × 1) **[10]**

QUESTION 2: CAVITY WALL CONSTRUCTION

2.1



Strip foundation	4
Concrete slab	2
Plaster	2
Screed	2
Hardcore	2
Ground level	2
DPC	2
Cavity	2
Wall tie	2

(20)

2.2 Minimum 50 mm✓ and maximum 100 mm.✓ (2)

2.3 Brick force makes masonry walls stronger and less likely to crack because of the natural movements of the wall or normal movements caused by either temperature changes or the natural expansion or contraction of brick or concrete products. (2)

2.4
- Special treatment is required at door and window openings where the inside and the outside walls meet.
- The cavity must be sealed at roof level, which may increase labour costs.
- The weep holes left in the face of the wall will attract bees.
- The internal dimensions of the rooms are reduced.
- It is expensive to build or construct it. (Any 4 × 1) (4)

2.5 With dripstones facing downwards or sloped towards the outside leaf (2)
[30]

QUESTION 3: BASIC MASONRY WALL CONSTRUCTION WITH HOLLOW UNITS

3.1 Jointing is the finish given to both the vertical and the horizontal joints in brickwork, irrespective of whether the walls are made of brick, block, solid or cavity construction.
Pointing is the finish given to joints by raking out to a depth of approximately 20 mm and filling in on the face with a hard-setting cement mortar. (2 × 2) (4)

3.2
- Flush struck
- Concave
- Vee
- Weather-struck
- Raked
- Struck (Any 5 × 1) (5)

3.3
- Strip foundation
- Raft foundation
- Cantilever foundation
- Pad foundation or column foundation
- Stepped foundation (Any 4 ×1) (4)

3.4
- Full unit✓ – most of the walls✓
- ¾ unit✓ – wall✓
- ½ unit✓ – intersections, returns✓
- ¼ unit✓ – door frames, window frames and lintels✓
- Sill unit✓ – window sills✓
- Lintel unit – lintels
- Sash unit✓ – Window frames✓
- Bond unit✓ – Lintels/ring beams✓ (Any 2 × 5) (10)

3.5 A gauge rod has a long, straight edge, marked at intervals equal to the thickness of the masonry unit plus the mortar joint, and it is used for checking and keeping masonry courses even.
A corner profile is a length of steel tubing of 50 mm × 50 mm and it is set up plumb at the corners of the walls to assist the mason in building the walls plumb and level.
(2 × 2) (4)

3.6 3.6.1 Where the wall turns a corner

3.6.2 A cross wall or intersecting wall is a wall that crosses or intersects a structural wall or outside wall.

3.6.3 A pipe through which a set of electrical wires can pass

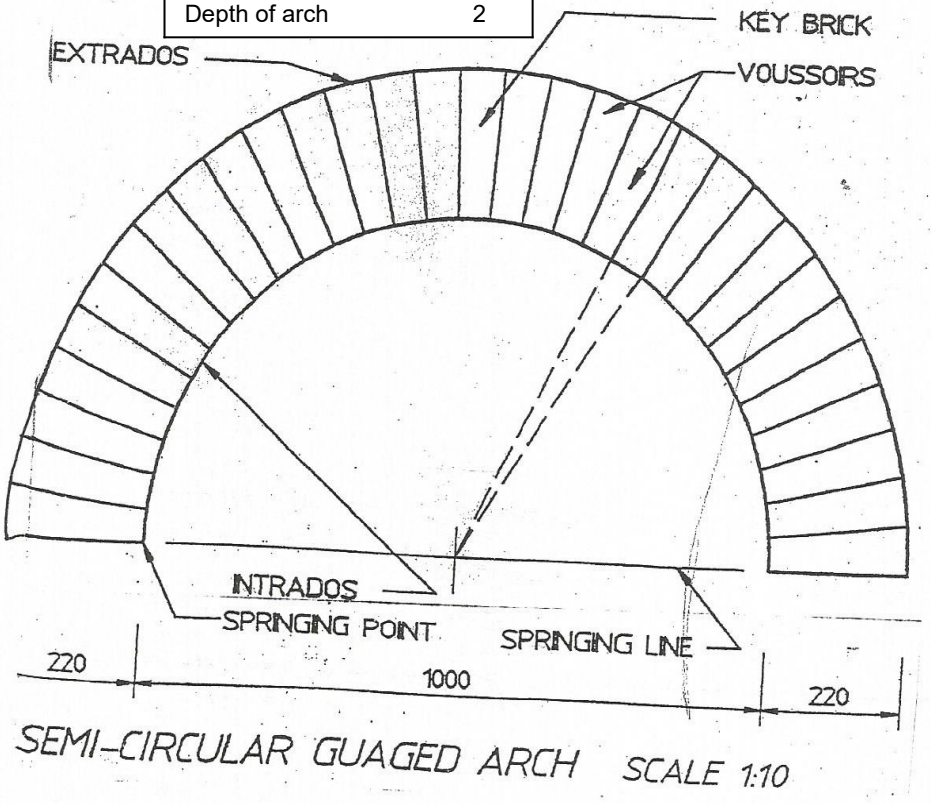
(3 × 1) (3)
[30]

QUESTION 4: ADVANCED MASONRY WALL CONSTRUCTION

4.1
- Use a raking tool to rake out the joints and to remove excess mortar.
- Then use a round rule or a square rule to profile joints.
- The mortar must still be thumbprint hard when you rake out and rule joints. (3)

4.2

Correctness	8
Span	2
Springing point & line	2
Extrados	2
Voussoirs	2
Intrados	2
Depth of arch	2



(20)

4.3
- Truss-type
- Ladder-type (2)

4.4
- Lightweight roof – 600 mm
- Heavyweight roof – 300 mm (2)

4.5
- The rise must be between 0,3 and 0,5 times the span.
- Solid masonry units must to be used.
- The arch ring must be constructed in either a header or a stretcher pattern.
- The arch-ring depth must not be less than 200 mm where the rise is between half or two thirds of the radius.
- The arch-ring depth must be not less than 300 mm where the rise is greater than two thirds but less than or equal to the radius. (Any 3 × 1) (3)

[30]

TOTAL: 100

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